
Lecture 7

ECE 4107 (Section A)

Telecommunication Engineering



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2.7 1000-line Blocking Exchange

- ☐ A blocking design is considered.
- ☐ Subscribers are identified by a three-digit number ranging from 000 to 999.
- ☐ Selectors used:
 - Preselector/Group Selector/Final Selector



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2.7 1000-line Blocking Exchange

- One uniselector for each subscriber to hunt a free Group selector (GS) in preselection stage.
 - Each uniselector has 24 outlets.
 - Each outlet of the uniselector is connected to a specific GS.
 - Corresponding outlets of all uniselectors are commoned to share GS.



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2.7 1000-line Blocking Exchange

- A number of two-motion selectors (GS) are employed for common use.
 - Each outlet is connected to a specific final selector (F/S).
 - Outlets of each vertical level of all GS's support a group of 100 subscribers.
 - Controlled by the first digit.
 - The number of GS can be changed to obtain good quality of services.



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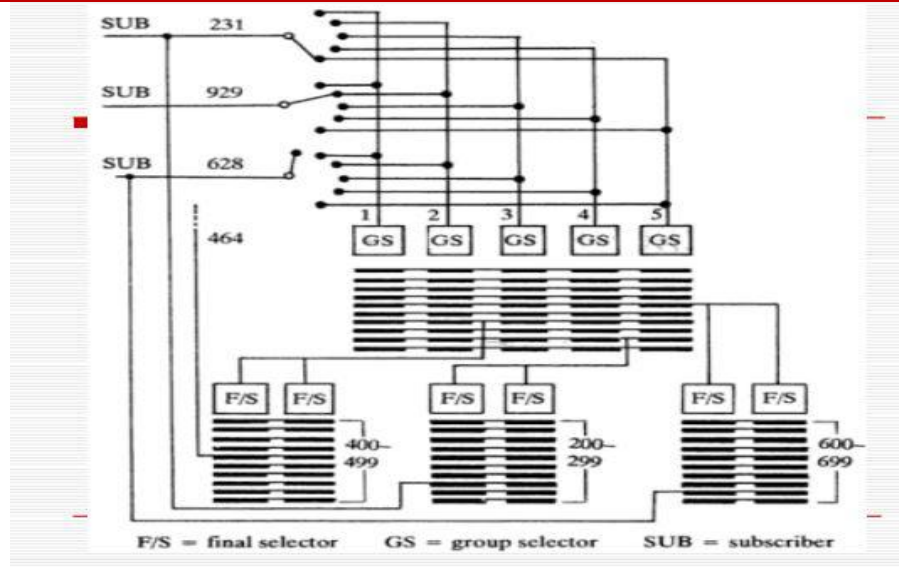
2.7 1000-line Blocking Exchange

- A number of two-motion selectors are employed as F/S.
- All F/S's are divided into 10 groups.
- Each group are connected to a specific level of GS and commoned in outlets to support 100 subscribers.
- The number of F/S can be changed to support more simultaneous calls.



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2.7 1000-line Blocking Exchange

□ Example 1

- In a 1000-line exchange, the number range 000-299 is allotted to business subscribers. Forty percent of these subscribers in each group of 100 are active during peak hours.
- The number range 300-999 is allotted to domestic connections. Ten percent of the domestic subscribers are active in each group at any time.
- Estimate the total number of final selectors required.???



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2.7 1000-line Blocking Exchange

□ Solution

- Number of simultaneous calls for business subscriber groups is equal to 20 per group;
- Number of simultaneous calls for domestic subscriber groups is 5 per group.
- Total number of final selectors required is: $3 \times 20 + 7 \times 5 = 95$



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2.7 1000-line Blocking Exchange

□ Example 2

- In example 1,
 - The probability of more than 40 percent business customers being active is 0.01.
 - The probability of more than 10 percent of the domestic customers being active is 0.05.
 - Assume that switching stages other than final selector stage are designed to be nonblocking.
- Estimate the blocking probability of the exchange. ???



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2.7 1000-line Blocking Exchange

□ Solution

- The exchange appears blocking in two cases:
 - A business customer is blocked when more than 40 percent business customers are active. The corresponding probability is 0.01.
 - A domestic customer is blocked when more than 10 percent domestic customers are active. The corresponding probability is 0.05.
- As a result, the blocking probability is

$$P_B = 0.05 + 0.01 = 0.06$$



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Design parameters for 1000-line exchange

- Consider 200 two-motion selector in second stage and also 200 for final stage.
- $S = 1000 + 400$, $SC = 200$, $K = 3$, $C = 1000 + 800 = 1800$
- $TC = 0.4$, $EUF = 0.57$, $CCI = 111.11$



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P_b

- Blocking Probability depends on traffic characteristics in each group
- Assume 20 final selectors in each group
- Probability of one call land on the a given block = $100/1000$
- Probability of two calls land on the same given block
 $= (100/1000) \times (99/998)$
- Probability of maximum 20 calls on the same block = $100/1000 \times 99/998 \times \dots \times 81/962$



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10,000-line exchange

- Four digits subscriber 0000.....9999
- Four stages switch network
- 2-sub stages in each group selectors stage
- When subscriber dial digits the first selector responds to first digit, second selector to the second digit and final selector to the last digit.
- To support 10000 subscriber we need a minimum 100 final selectors



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Find the Design Parameters (10000), consider 1000 simultaneous call

- $S = 10000 + 1000 + 1000 + 1000$, $SC = 1000$, $K = 4$, $TC = 0.2$
- $EUF = 0.3$, $C = 16000$, $CCC = 62.5$



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❑ Disadvantages of Strawger Switching

- Dependence on moving parts and contacts that are subject to wear and tear;
- Difficulties in maintenance and adjustments.

❑ Switching systems requiring less maintenance and adjustment after installation are expected.



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Crossbar Switch

Introduction

❑ Development of Crossbar Switching

- The first patent, USA, 1915.
- The Second patent, Sweden, 1919.
- The first Crossbar Switching system, AT&T, 1938.
- **Common Control** subsystems were first introduced in crossbar exchanges.



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3.1 Principles of Common Control

□ Uniform numbering

- The same number is dialed, no matter from which exchange the call originates.
- It is convenient for customer to make phone call to a specific party any where.
- It is difficult to implement uniform numbering in **direct control** switching system unless a 'Director' is available.



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A Multi-exchange Network

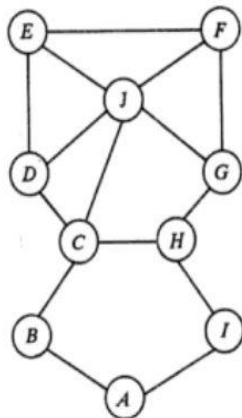


Fig. 3.1 A multiexchange network.

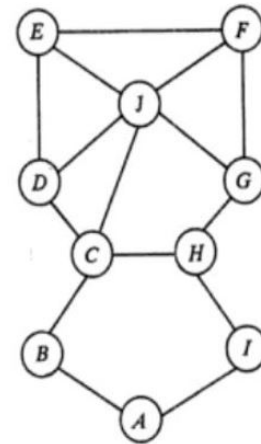
From exchange	Outlet	To exchange
A	01	B
A	02	I
B	04	C
C	03	J
I	05	H
H	01	G
G	02	F
J	01	F



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- A call from A to F
 - Called subscriber: 1457
 - Route 1: A-B-C-J-F
dialing sequence: 01-04-03-01-1457
 - Route 2: A-I-H-G-F
dialing sequence: 02-05-01-02-1457



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A Multi-exchange Network

- Difficulties faced by direct control
 - Identification number of a subscriber is **route dependent**;
 - A user must have knowledge of the **topology** of the network and the **outlet assignments** in each exchange.
 - Depending on from which exchange the call originates, **the number and its size vary** for the same called subscriber.



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A Multi-exchange Network

- How to overcome these difficulties in a Strawger exchange?
 - The routing is done by the exchange instead of dialling.
 - A uniform numbering scheme is used.
- Uniform Numbering scheme
 - Exchange identifier + Subscriber line identifier



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Common control

- Fundamental features
 - A user is assigned a **logical number** which is independent of the **physical line number** used to establish a connection to him.
 - The logical address is translated to actual physical address for connection establishment by an **address translation mechanism**.
 - Call processing takes place independent of the switching network.



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Essential Features of Common Control

- Exchange must have capable of receive/store the digit dial
- Translate the exchange identifier into routine digits
- Transmit the routine and subscriber line identifier digits to switch network



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CT1

Date : 24.02.2026

Time : 11 am

Syllabus : Six different design method (100 line+1000 line)



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